JAN-I-7

31 May 1944

JOINT ARMY-NAVY SPECIFICATION

INSULATORS, GLASS-BONDED-MICA, RADIO

Army Number
71-4933

Navy Number
16-I-7

This specification was approved by the War Department and the Navy Department for use of procurement services of the Army and Navy.

A. APPLICABLE SPECIFICATIONS.

A-1. The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

JOINT ARMY-NAVY SPECIFICATION

JAN-I-10.—Insulating material, ceramic, radio, class L.

NAVY DEPARTMENT SPECIFICATIONS

General Specifications for Inspection of Material.[1]
39P16—General specification for packaging and packing for overseas shipment.

B. TYPE.

B-1. Glass-bonded mica radio insulators covered by this specification shall be of but one type, fabricated by machining to the desired shape.

C. MATERIAL AND WORKMANSHIP.

C-1. *Material.*—

C-1a. The material for each part shall be as hereinafter specified. An adequate material shall be used when a definite material is not designated.

C-1b. Glass-bonded mica used in the construction of insulators shall be an approved grade conforming to the requirements of Joint Army-Navy Specification JAN-I-10.

C-2. *Substitution of material.*—If the manufacturer desires to substitute for a specified or previously approved material or fabricated

[1] Applicable only to Navy purchases.

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part, he shall submit to the bureau or agency concerned with his bid or sample a statement to that effect, describing the proposed substitution, together with evidence to substantiate his claim that such substitute is adequate. At the discretion of the bureau or agency concerned, test samples may be required to prove the suitability of the proposed substitute.

C-3. *Standard screw threads.*—Where practicable, screw threads for bolts, machine screws, nuts, and tapped holes shall be in strict accordance with the Screw-Thread Standards for Federal Services, Handbook H-28. Unless otherwise specified, screws smaller than size 6-32 will not be permitted.

C-4. *Workmanship.*—All parts shall be manufactured and processed in a careful and workmanlike manner, in accordance with the best design and practice for use in tropical and cold climates, and must meet the design, performance, and test requirements of this specification and of the applicable detail drawings.

D. GENERAL REQUIREMENTS.

D-1. *Definitions.*—

D-1a. *Holes.*—

D-1a (1). *Mounting holes.*—Mounting holes are defined as those holes in the insulator which are used as points of support for the insulator or for other members of an assembly.

D-1a (2). *Other holes.*—Other holes are defined as all holes other than mounting holes, such as winding or tap-off holes in coil forms, lead holes, perforations for ventilation or for weight reduction purposes, etc.

D-1b. *Blisters.*—Small bulges or lumps on the surface of a ceramic piece are known as blisters. The term does not necessarily indicate the presence of a cavity under the surface.

E. DETAIL REQUIREMENTS.

E-1. Insulators covered by this specification shall be manufactured in accordance with the applicable drawings (see par. H-2a).

E-2. *Fittings.*—Insulators shall be furnished with such fittings as specified on the applicable drawing or in the contract or order (see par. H-2b). The methods employed for securing the fittings shall be of such nature as to insure that the completed insulator will meet all specified tests.

E-3. *Condition of surfaces.*—Surfaces which are chipped or have other imperfections to the extent that the life and usefulness of the insulator are impared shall be cause for rejection. Rough or irregular edges caused by minute chips or nicks are not considered as impairing the life and usefulness of the insulator unless sharp edges are specified on the applicable drawing. The following information shall be used in determining the significance of all surface imperfections:

E-3a. *Chips.*—Insulators having surfaces which are chipped shall be rejected in the event that such imperfections result in:

(1) Surfaces surrounding mounting holes which will not withstand the stress of mounting or will not provide secure mounting.

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(2) Bearing surfaces which will not withstand bearing pressure or friction of associated assembly details.
(3) Surfaces of grooves or other indentations provided for carrying coil windings or other wiring which will not provide adequate support for winding or which will damage the wire.
(4) Sealing surfaces which will not provide sufficient support for materials used in sealing inserts or associated assembly details to the insulator.

NOTE.—It is deemed impractical to set up even general standards of quality for size and quantity of chips inasmuch as either must be judged in conjunction with the other. Since both size and application of glass-bonded mica parts are subject to wide variation from one piece to another, it is believed that questions of size and quantity of chips should be considered on a common-sense basis rather than by any empirical formula covering "number of chips per square or linear inch" or "loss in volume."

E–3b. *Blisters.*—The surfaces of glass-bonded mica insulators shall be free from bulges or blisters.

E–3c. *Cracks.*—Insulators shall be free from cracks.

E–3d. *Surface color.*—Variation in surface color shall not in itself constitute basis for rejection.

E–4. *Surface treatment.*—Where a surface treatment is desired, it shall be as specified in the contract or order (see par. H–2c).

E–5. *Tolerances.*—In general, and unless otherwise specified, the following tolerances shall apply to all dimensions of insulators:

E–5a. *Over-all dimensions (length, width, and thickness).*—The tolerance for any over-all dimension shall be as shown in table I.

TABLE I

Length	Tolerances (±)		
	Length	Width	Thickness
Inches	*Inch*	*Inch*	*Inch*
Up to 16	1/64	1/64	1/64
16 or greater	1/64	0.020	1/64

E–5b. *Hole spacing.*—A tolerance of ± 0.007 inch will be permitted in dimensions for distances between hole centers and distances from hole centers to reference edges or reference lines (see fig. 2 of Appendix).

E–5c. *Hole size.*—Standard tolerances for all hole diameters shall be ± 0.005 inch for all diameters up to 0.5 inch and ± 1 percent of the hole diameter for all diameters 0.5 inch and greater.

E–5d. *Angular dimensions.*—Where angles or V cuts are required in the edges of flat pieces, the tolerance on such angular dimensions shall be ± 1 degree (see fig. 2 of Appendix).

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E–5e. *Flatness.*—Where flatness limits are not indicated on the drawing or in the contract or order, the limit shall be 0.0015 inch, per inch of length for any dimension.

E–5f. *Diameter of rods.*—A tolerance of ± 0.005 inch will be permitted in the diameter of all round rods.

E–6. *Tapped holes.*—Threads in tapped holes shall conform to class 1 fit, as outlined in Screw-Thread Standards for Federal Services, Handbook H–28, except when otherwise specified on the applicable drawing.

E–7. *Test requirements.*—Where test requirements are included on the applicable drawing or in the contract or order (see par. H–2d), insulators shall be capable of complying with the indicated tests (see par. F–3b).

F. METHODS OF SAMPLING, INSPECTION, AND TESTS.

F–1. *Test equipment and inspection facilities.*—The insulator manufacturer shall furnish all necessary facilities and equipment for making all inspection tests required and shall carry out all tests under the supervision of the Government inspector. The test equipment shall be adequate in quantity to enable the inspection to keep up with production. The Government inspector shall ascertain that the equipment for making tests meets all the requirements of this specification. The insulator manufacturer shall correct all deviations from this specification pointed out by the Government inspector. Wherever practicable, the inspection of insulators shall be at the plant of the insulator manufacturer.

F–2. *Special gages.*—If special gages for inspection of parts are to be used in the determination of acceptability of the insulators, specifications for such gages or a set of gages themselves, if practicable, shall be furnished to the insulator manufacturer by the bureau or agency concerned.

F–3. *Production tests.*—

F–3a. *Visual inspection.*—

F–3a (1). Each insulator shall be inspected to verify that the workmanship and specified physical dimensions and tolerances are satisfactory. Screw gages or screws conforming to "class 1 fit" shall be used for the inspection of tapped holes.

F–3a (2). Insulators failing to meet such inspection shall be rejected unless it can be shown to the satisfaction of the Government inspector that the insulator in question can be corrected with respect to the substandard features, in which case it shall be corrected and reinspected.

F–3b. *Other tests.*—

F–3b (1). Where other tests, such as mechanical load or high-potential tests are specified on drawings or in the contract or order, insulators shall be selected by the Government inspector and subjected to the applicable tests. Unless otherwise specified, in the contract or order, the number of samples and the allowable maximum number of failures shall be as shown in table II.

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TABLE II

Quantity of insulators on order	Number of samples for test	Maximum number of failures permitted
0 to 50	0	----------
51 to 100	4	0
101 to 500	6	1
501 to 5,000	12	[1] 3
Each additional 5,000	12	[1] 3

[1] Total of 3 failures, but not more than 2 failures permitted for any one test.

F-3b (2). If the number of failures exceeds the requirements of table II, a retest shall be made. If the number of failures of the same type still exceeds the maximum permissible, this shall not necessarily be cause for complete rejection. If the failures on the retest exceed 2 percent of the quantity inspected, then each insulator on the order shall be subjected to the applicable tests. Insulators meeting the test requirements shall be accepted, and a sufficient number of acceptable insulators shall be processed and tested to replace those that failed.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT.

G-1. *Army.*—Shipments of the subject equipment shall be packaged, packed, and marked, as specified in the bid request, contract, or basic equipment specification (see par. H-2).

G-2. *Navy.*—

G-2a. *Packaging.*—Unless otherwise specified, the subject commodity shall be packaged in uniform carrying boxes of two-piece telescopic style, made of 0.020-inch bleached manila lined chip or better. A uniform count per package (not to exceed one hundred) shall be maintained. When two or more adjacent rows of subject commodity are packaged, separator pads made of 0.020 bleached manila lined chip or better shall be used between rows to protect insulator edges.

G-2b. *Packing.*—

G-2b (1). *For domestic shipment.*—Unless otherwise specified, the subject commodity, packaged as described in paragraph G-2a, shall be packed in Style RSC corrugated or solid fiberboard boxes conforming to the following requirements:

Gross weight (max.)	Average bursting strength (min.)	Sum of inside dimensions (max.)
Pounds	*Pounds*	*Inches*
40	200	60
65	275	75
90	350	90

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In order to permit inspection of the contents, boxes shall be securely sealed in such a manner as to permit opening and subsequent reclosure without damage to the upper flaps.

G–2b (2). *For overseas shipment.*—Unless otherwise specified, the subject commodity, packaged as described in paragraph G–2a, shall be packed in cleated plywood, nailed wood or weatherproof fiberboard boxes, conforming to the requirements of sections V, VI and VIII, respectively, of Navy Department Specification 39P16. The gross weight shall not exceed 75 pounds.

G–2c. *Marking.*—

G–2c (1). *Packages.*—Unless otherwise specified, packages shall be conspicuously marked on each of two adjacent sides in black with the following information completed:

ARTICLE_____
CONTRACTOR_____
MANUFACTURER_____
TYPE NUMBER_____
STOCK NUMBER_____
ORDER NUMBER_____
QUANTITY_____

G–2c (2). *Shipping containers.*—Unless otherwise specified, shipping containers shall be conspicuously marked with the name of the article, type designation, purchaser's stock number, quantity contained therein as defined by the contract or order under which shipment is made, the name of the contractor, the number of the contract or order, the gross weight, and the words "FRAGILE" and "HANDLE WITH CARE." In addition to the foregoing, marking shall conform to the requirements of the latest issue of the Navy Shipment Marking Handbook.

H. NOTES.

H–1. Requests, requisitions, schedules, and contracts or orders should contain the title of the specification, the number, and date.

H–2. Requests, requisitions, schedules, and contracts or orders should specify the following:

(a) Applicable drawings.
(b) Fittings desired.
(c) Surface treatment, where required (see par. E–4).
(d) Special tests, where required (see par. E–7).
(e) Packaging, packing, and marking requirements for Army purchases (see par. G–1).

H–3. This specification has been editorially rearranged but is identical in all essential technical respects to the July 8, 1943 issue of the American War Standard C75.6–1943. Reference to American War Standard C75.1–1943 has been deleted and Joint Army-Navy Specification JAN-I-10, which is identical in all essential technical respects, is referenced. Paragraphs on approval of material have been deleted as approval is covered in the material specification.

H–4. This specification has been prepared through the coordinated efforts of representatives of industry and the War and Navy Depart-

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ments. This specification should be used in the design of new equipment and, insofar as is practicable, replacement parts should also conform to the requirements specified herein.

H–5. This specification is not intended to be restrictive with respect to the details of construction, except where such details are specified. Alternate forms of construction will be considered provided the bid clearly describes the specific points in which the construction proposed differs from that described herein, and provided further, that all performance tests and other requirements of this specification are met. When exceptions are not clearly described and indicated as such, it will be assumed that bidders are offering insulators in strict accordance with the requirements of this specification.

H–6. Matters relating to the use of components furnished in accordance with this specification for use in connection with the design, production, and procurement of all Naval Radio, Radar, and Underwater Equipment on the Electronics Precedence List shall be referred to the Bureau of Ships Radio Division, Navy Department, Washington 25, D. C.

H–7. Copies of the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C.

H–8. Copies of Joint Army-Navy specifications (required for Army purchases) may be obtained, as indicated in the "Index of United States Army and Federal Specifications Used by the War Department." Copies of this Index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Agencies within the War Department will obtain copies of Joint Army-Navy specifications through established War Department channels. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–9. Copies of Joint Army-Navy specifications (required for Navy purchases) and Navy Department specifications may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Depot, Bayonne, N. J. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

NOTICE.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation the United States Government thereby incurs no responsibility or any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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APPENDIX

DESIGN CRITERIA

This section on design criteria was developed to be used as a reference for engineers and chief draftsmen in designing glass-bonded mica into their equipment. The data compiled represent that which is considered good design practice and does not necessarily imply that closer tolerances on any important dimensions or special designs cannot be produced by special handling such as molding with or without metal inserts, hot punching, or hot forming. It is definitely recommended that the following material be carefully considered by engineers and draftsmen in designing machined glass-bonded mica items, thus increasing production by reducing the possibility of waste, delay, and assembly difficulties. In order to insure the most desirable design and to facilitate procurement thereof at minimum cost and production time, it is recommended that designers consult the insulator manufacturer or fabricator concerning specific design details applicable to the insulator under consideration.

1. *Preferred shapes.*—To provide maximum production, reduce fabrication time, and reduce costs, it is recommended that rectangular parts be used where possible and that irregular or curved sections be avoided.

2. *Tapped holes.*—Although holes in glass-bonded mica insulation can be tapped, it is recommended that such holes be avoided if possible. If tapped holes are required, they should be slightly countersunk to prevent chipping.

2a. Bottoming of tapped or blind holes should be avoided.

2b. Threads in tapped holes should conform to class 1 fit as outlined in Screw-Thread Standards for Federal Services, Handbook H–28.

3. *Counterboring.*—Although it is possible to counterbore glass-bonded mica, it is not always possible to secure sharp edges around the counterbore.

4. *Thickness of flat parts.*—Flat plates or sheets from which most parts are fabricated are available in standard thicknesses of 1/8, 3/16, 1/4, 3/8, 1/2, 5/8, 3/4, 7/8, and 1 inch. In the design of flat parts, standard thicknesses should be specified wherever possible. It should be noted that the nominal minimum thickness of flat parts is 1/8 inch. Although thicknesses less than 1/8 inch can be produced, special grinding operations involving high cost and increase in production time are required. In any case, the thickness should be not less than 3/32 inch where the maximum dimension (length or width) of the part is 3 inches. For dimensions above 3 inches, the minimum thickness should be 1/8 inch.

5. *Round rods.*—Round rods are available in standard dimensions of 1/4, 3/8, 1/2, 5/8, 3/4, 7/8, and 1 inch. In the design of parts to be machined from rod stock, such as bushings, washers, or spacers, standard diameters should be specified wherever possible (see par. 12g of Appendix).

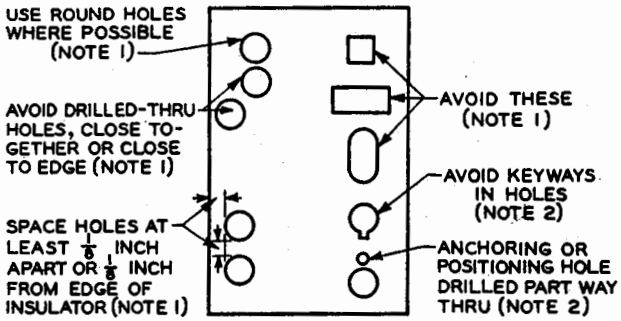
6. *Corners.*—On flat plate designs, right-angle outside corners should be specified where possible to avoid additional time-consuming machin-

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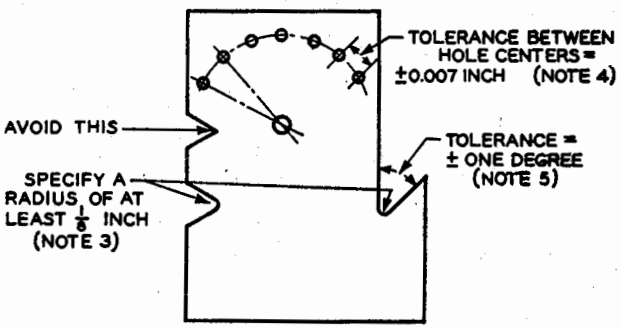
ing operations. Where it is essential that sharp outside corners be eliminated, a chamfer should be used in place of a definite radius, as the latter requires the use of special jigs (see fig. 8).

7. *Angles.*—Where angles or V cuts are required in the edges of flat pieces, a radius of not less than ⅛ inch should be specified at the apex of the angle or V cut (see fig. 2).

8. *Holes.*—Round holes should be specified in preference to square, rectangular, or elongated holes as the latter involve costly, time-consuming machine operations. The distance between edges of holes and between an edge of the insulator and the edge of a hole should be not less than ⅛ inch (see fig. 1).

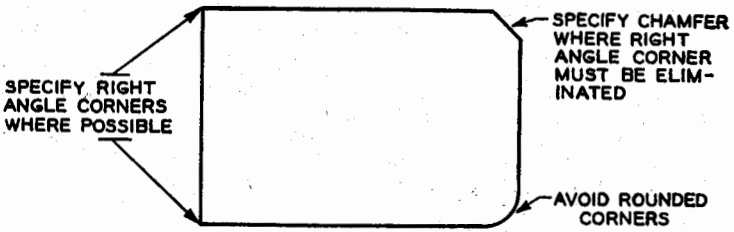


HOLES AND HOLE SPACING
FIG. 1

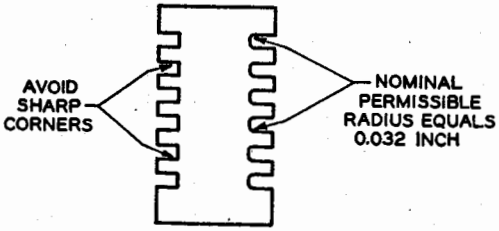


ANGLES AND ANGULAR DIMENSIONS
FIG. 2

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CORNERS (NOTE 6)
FIG. 3



SLOTTED BARS (NOTE 7)
FIG. 4

NOTE 1.—See Appendix, paragraph 8.
NOTE 2.—See Appendix, paragraph 9.
NOTE 3.—See Appendix, paragraph 7.
NOTE 4.—See paragraph E–5b of specification; Appendix, paragraphs 12b and 12e.
NOTE 5.—See paragraph E–5d of specification; Appendix, paragraph 12d.
NOTE 6.—See Appendix, paragraph 6.
NOTE 7.—See Appendix, paragraph 11.

9. *Key ways in holes.*—The machining of key ways, slots, or notches in holes is costly and time-consuming since milling, grinding, or filing operations are required. Alternative methods of anchoring metal parts should be considered by the designer. A small hole close to the main mounting hole may be used to receive an anchoring or positioning pin. Such small holes are usually drilled partially through the insulator. If a key way is essential, a radius of 0.032 inch should be allowed on the inside corners (see fig. 1).

10. *Bushings.*—The wall thickness of bushings should be not less than as shown in table I.

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TABLE I

Maximum bushing length	Outside diameter	Minimum permissible wall thickness
Inch	*Inch*	*Inch*
¼	Up to ⅜	1/16
1	Up to ½	⅛
1	½ to ¾	3/16
1	¾ to 1	¼

11. *Slotted bars.*—A radius of 0.032 inch should be allowed on the inside corners of slots cut in the edge of parts (see fig. 4).

12. *Tolerances.*—In general, and unless otherwise specified, the following tolerances should apply to all dimensions of insulators:

12a. *Over-all dimensions (length, width, and thickness).*—The tolerance for any over-all dimension should be in accordance with table II.

TABLE II

Length	Tolerances (±)		
	Length	Width	Thickness
Inches	*Inch*	*Inch*	*Inch*
Up to 16	1/64	1/64	1/64
16 or greater	1/64	0.020	1/64

12b. *Hole spacing.*—A tolerance of ± 0.007 inch should apply to dimensions for distances between hole centers and distances from hole centers to reference edges or reference lines. Although it is possible to fabricate to closer tolerances, such tolerances involve special fabricating tools and technique and should therefore be avoided unless essential (see fig. 2).

12c. *Hole size.*—Standard tolerances for all hole diameters should be ± 0.005 inch for all diameters up to 0.5 inch, then ± 1 percent of hole diameter for all diameters 0.5 inch and greater.

12d. *Angular dimensions.*—Where angles or V cuts are required in the edges of flat pieces, the tolerance on such angular dimensions should be ± 1 degree (see fig. 2).

12e. Where angular dimensions are specified for the layout of holes, the tolerance on dimensions between hole centers should be ± 0.007 inch. It is recommended that the chord distance between hole centers be indicated on drawings to assist in the layout of hole centers (see fig. 2).

12f. *Flatness.*—Where flatness limits are not indicated on the drawing or specification, the limit should be 0.0015 inch, per inch of length for any dimension.

12g. *Diameter of rods.*—A tolerance of ± 0.005 inch should apply to the diameter of all round rods.

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